

Wholesale Prices in India

1. Project Overview:

This project focuses on analyzing wholesale commodity prices in India using Excel & Power BI. The dataset contains historical price information for various commodities such as Milk, Rice, Wheat, Sugar, and others. Each record includes the date and different price measures, including Average Price, Maximum Price, Minimum Price, and Modal Price.

2. Project Objective:

The main objective of this project is to analyze wholesale commodity prices in India and gain meaningful insights into price trends and fluctuations over time. By using Power BI, the project aims to transform raw pricing data into interactive dashboards that help users understand the behavior of different commodities based on their Average Price, Maximum Price, Minimum Price, and Modal Price.

3. Data source:

Dataset source: India data portal

Timeline: 2015 - 2024

Data Size: 10,000+ rows and 7 columns

4. Data Cleaning:

Before importing the wholesale prices dataset into Excel and Microsoft Power BI, the raw data was cleaned and standardized to ensure accurate analysis and reporting.

4.1. Removed Duplicate Records

- Used **Data → Remove Duplicates** in Excel.
- Checked the **ID** column to ensure each record was unique.

- Eliminated repeated rows to maintain data integrity.

4.2. Corrected Date Format

- Converted the **Date** column into a proper Date format (DD/MM/YYYY).
- Ensured Excel recognized all values as valid da

4.3. Standardized Commodity Names

- Used TRIM() to remove extra spaces.
- Used PROPER() to apply consistent capitalization.
- Corrected inconsistent spellings in the **Commodity** column.

4. 4. Verified Numeric Columns

The following columns were checked and converted to numeric format:

- Average_price
- Maximum_price
- Minimum_price
- Modal_price

This ensured that calculations and visualizations would work correctly.

4.5. Handled Missing Values

- Identified blank or null cells.
- Replaced missing values where appropriate

4.6 . Saved Clean Dataset

- Saved the cleaned file in .xlsx or .csv format.
- Imported the cleaned dataset into Power BI for further analysis.

4.7.Final Cleaned Columns

- ID

- Date
- Commodity
- Average_price
- Maximum_price
- Minimum_price
- Modal_price

5.Tools Used:

- Excel (Power Query)
- Power BI

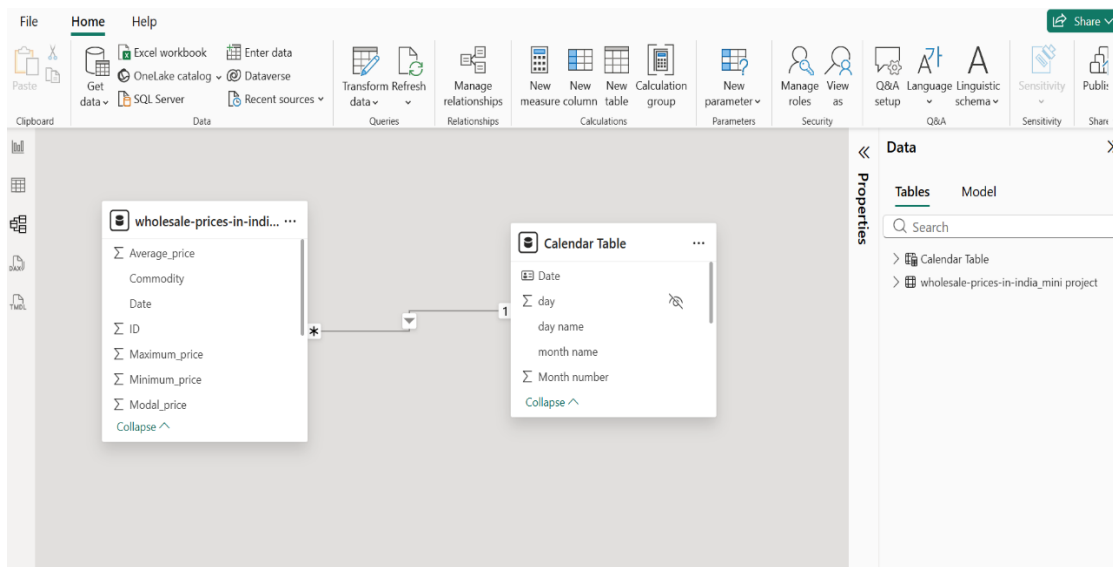
6.Data Transformation:

To enhance analysis, the following transformations were applied:

- Created new Calendar tables

7.Data Modelling:

To perform time-based analysis, a dedicated Calendar Table was generated and related to the main wholesale price table using the Date field. The relationship was established as one-to-many, where the Calendar Table contains unique dates and the main table contains multiple records for each date. This structured data model allows for advanced analysis by day, month, and year, and ensures better performance and more accurate DAX calculations throughout the report.



8.Exploratory Data Analysis (EDA) & Visualization:

- Performed statistical analysis on wholesale price data, including Average Price, Maximum Price, Minimum Price, and Modal Price.
- Conducted univariate analysis to understand the distribution of prices across different commodities.
- Performed bivariate analysis to compare price variations over time and across commodities.
- Conducted multivariate analysis using Commodity, Date, and multiple price measures to identify patterns and trends.
- Built interactive dashboards and visualizations in Power BI to present key insights effectively.

8.1 Features Included:

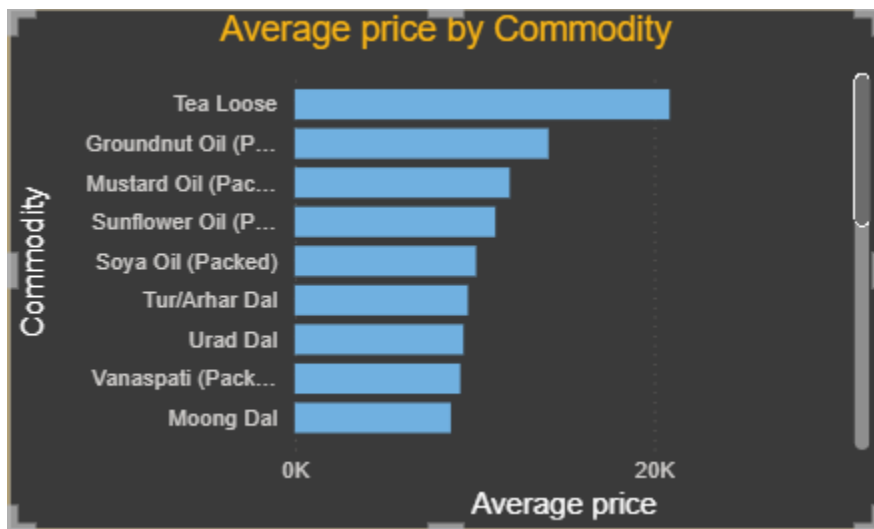
- Filters (Slicers) for Commodity, Year, and Month.
- Interactive visuals such as line charts, bar charts, cards, and tables.



9. Key Insights (SMART + Analytics Approach):

Types of Analytics Used

9.1 Bar Chart - Average Price by Commodity



Descriptive Analytics (What happened)

The bar chart shows the average prices of commodities, where Tea Loose has the highest average price and Moong Dal has the lowest, making it easy to compare costs across items.

Diagnostic Analytics (Why it happened)

The differences in average prices happened because of factors like production costs, import dependency for oils, and seasonal supply variations for pulses.

Predictive Analytics (What will happen)

Based on past averages, predictive analytics can forecast that commodities like oils will likely remain expensive, while pulses may continue to fluctuate depending on crop yields and demand.

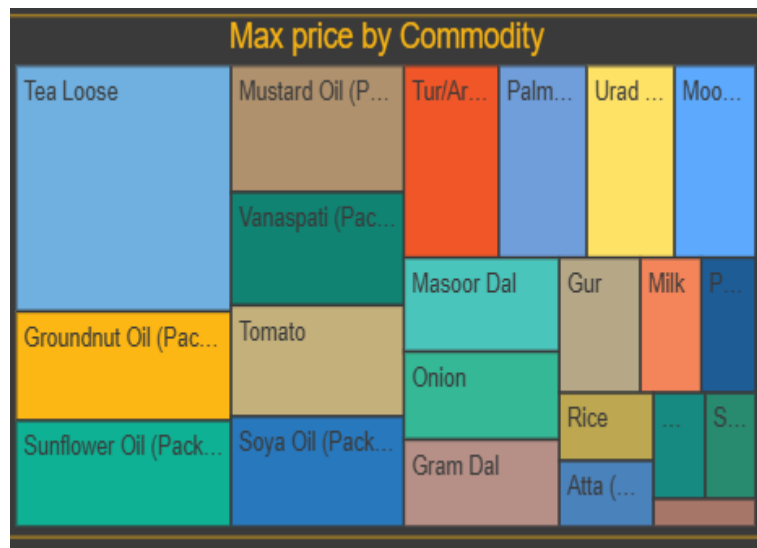
Prescriptive Analytics (What should be done)

Prescriptive analytics suggests actions such as diversifying suppliers, encouraging local production, or adjusting procurement strategies to manage high-cost commodities and stabilize future prices.

Insight

The bar chart highlights that Tea Loose consistently has the highest average price, while pulses like Moong Dal remain more affordable. Oils show higher costs due to external market factors, making them critical for cost-control strategies in procurement and planning.

9.2 Tree map - Max Price by Commodity



Descriptive Analytics (What happened)

The treemap shows the maximum prices of different commodities, where larger rectangles represent items like edible oils and pulses that reached higher price levels compared to staples such as rice, wheat, and sugar.

Diagnostic Analytics (Why it happened)

The differences in maximum prices happened because of factors like import dependency for oils, seasonal supply shortages for vegetables, and government policies that kept staples more stable.

Predictive Analytics (What will happen)

Based on past patterns, predictive analytics can forecast that seasonal items like onion and tomato will continue to show sharp price fluctuations, while edible oils may remain high if global supply chain pressures persist.

Prescriptive Analytics (What should be done)

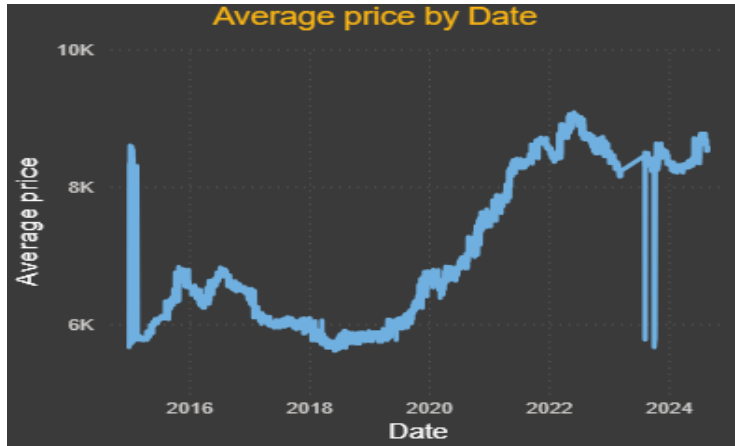
Prescriptive analytics suggests actions such as diversifying suppliers, improving storage facilities, and adjusting import strategies to manage commodities with high volatility and stabilize future prices.

Insight

The treemap highlights that edible oils dominate maximum price levels due to external market dependencies, while staples remain relatively stable. Seasonal

vegetables, however, are expected to stay volatile, making them critical for proactive planning and decision-making.

9.3 Line Chart - Average Price by Date



Descriptive Analytics (What happened)

The line chart shows how average commodity prices have changed over time, with peaks around 2021–2022 and dips around 2016 and 2024.

Diagnostic Analytics (Why it happened)

The price spikes and drops can be explained by factors such as supply shortages, seasonal demand, inflation, and government policies that influenced the market during those years.

Predictive Analytics (What will happen)

Based on past patterns, predictive analytics can forecast future average prices, showing likely increases during seasonal shortages and possible declines when supply improves.

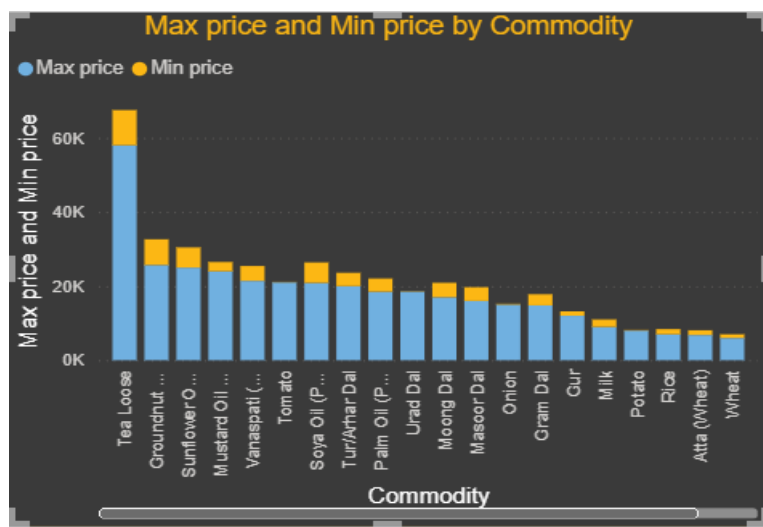
Prescriptive Analytics (What should be done)

Prescriptive analytics suggests actions like diversifying suppliers, improving storage, or adjusting imports to manage future price fluctuations and stabilize costs.

Insight

The line chart reveals that average prices are cyclical, with sharp rises during certain years. By combining descriptive, diagnostic, predictive, and prescriptive analytics, I can understand past trends, explain causes, anticipate future changes, and take proactive steps to manage commodity price risks.

9.4 Clustered Column Chart - Max Price and Min Price by Commodity



Descriptive Analytics (What happened)

The chart shows the maximum and minimum prices of commodities, where items like oils and pulses have wide price ranges, while staples such as rice and wheat remain more stable.

Diagnostic Analytics (Why it happened)

The differences between max and min prices happened because of supply chain issues, seasonal demand, import dependency for oils, and government policies that kept staple prices more controlled.

Predictive Analytics (What will happen)

Based on past ranges, predictive analytics can forecast that commodities with high volatility, like onions and potatoes, will continue to fluctuate, while staples will likely remain steady.

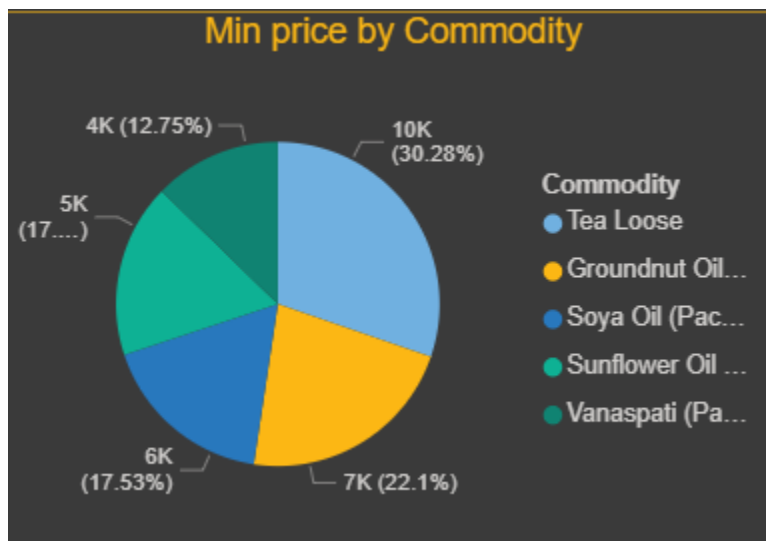
Prescriptive Analytics (What should be done)

Prescriptive analytics suggests actions such as building buffer stocks, diversifying suppliers, and improving storage facilities to reduce volatility and stabilize prices of sensitive commodities.

Insight

The chart highlights that commodities like oils and vegetables show large gaps between maximum and minimum prices, signaling high volatility. Staples remain relatively stable, making them less risky. This insight helps prioritize risk management strategies for volatile items while maintaining steady supply for essentials.

9.5 Pie Chart - Min Price by Commodity



Descriptive Analytics (What happened)

The pie chart shows the minimum prices of commodities, where Vanaspati has the largest share at 10K and Tea Loose has the smallest at 4K, making it easy to compare their relative costs.

Diagnostic Analytics (Why it happened)

The differences in minimum prices happened because of factors like production costs, import dependency for oils, and government policies that influenced price floors for certain commodities.

Predictive Analytics (What will happen)

Based on past minimum prices, predictive analytics can forecast that oils will likely remain higher, while commodities like Tea Loose may continue to stay at the lower end of the price range.

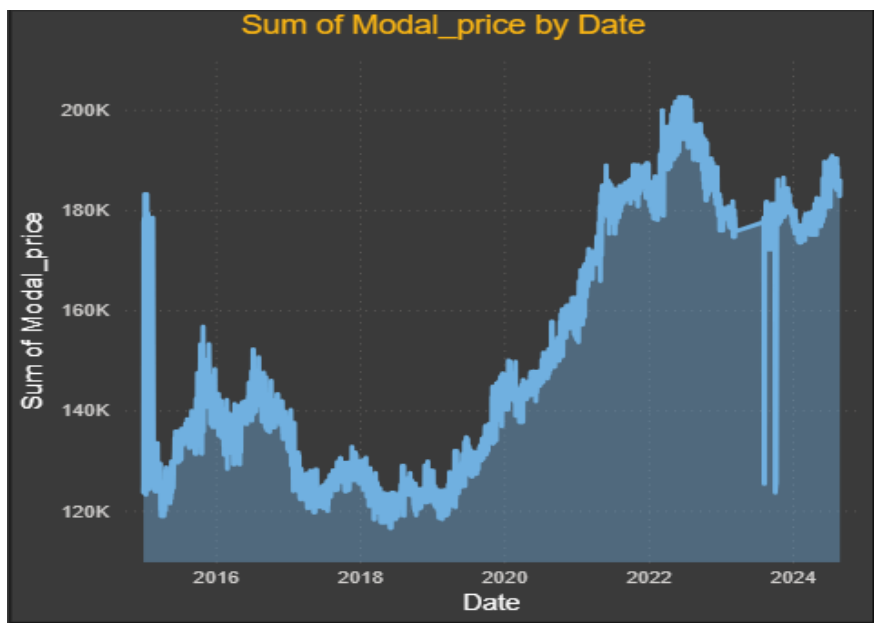
Prescriptive Analytics (What should be done)

Prescriptive analytics suggests actions such as encouraging local production, diversifying suppliers, and improving procurement strategies to manage commodities with higher minimum prices and reduce overall costs.

Insight

The pie chart highlights that Vanaspati and Sunflower Oil dominate minimum price levels, while Tea Loose remains the lowest. This insight shows which commodities are consistently costlier even at their minimum, guiding better procurement and cost-control decisions.

9.6 Area Chart - Sum of Modal Price by Date



Descriptive Analytics (What happened)

The area chart shows how the sum of modal prices changed over time, with peaks around 2016 and 2022 and a sharp drop near 2023.

Diagnostic Analytics (Why it happened)

These fluctuations happened because of factors like seasonal supply changes, inflationary pressures, global market disruptions, and government policies that influenced commodity prices during those years.

Predictive Analytics (What will happen)

Based on past patterns, predictive analytics can forecast that prices may rise again during seasonal shortages or global supply chain issues, while dips may occur when production improves or policies stabilize the market.

Prescriptive Analytics (What should be done)

Prescriptive analytics suggests actions such as building buffer stocks, diversifying imports, and improving storage facilities to manage future volatility and stabilize commodity prices.

Insight

The area chart highlights that modal prices are cyclical, with sharp rises and sudden drops across years. Understanding these patterns helps anticipate future risks and guides proactive strategies to stabilize markets and reduce price shocks.

10.Conclusion:

Wholesale Price in India dashboard brings together multiple visualizations—bar charts, tree maps, line charts, clustered columns, pie charts, and area charts—into one cohesive view. Here's a clear conclusion in your own words:

This dashboard provides a complete picture of commodity prices in India by showing averages, maximums, minimums, and modal price trends over time. It helps me understand past price behavior (descriptive), identify reasons for fluctuations (diagnostic), forecast future movements (predictive), and decide on actions to stabilize costs (prescriptive). Overall, the dashboard highlights which commodities are consistently high-priced, which are volatile, and how prices have shifted across years, guiding better planning, procurement, and policy decisions